frontiers in Marine Science

TYPE Original Research
PUBLISHED 24 July 2026
DOI 10.3389/fmars.2026.1885615

Check for updates

OPEN ACCESS

EDITED BY
Varenyam Achal,
Guangdong Technion-Israel Institute of Technology (GTIIT), China

REVIEWED BY
Gao Xiaoqian,
Shandong University of Science and Technology, China
Guojun Wu,
Chinese Academy of Sciences (CAS), China

*CORRESPONDENCE
Huayun Yu
397908756@qq.com

RECEIVED 19 May 2026
REVISED 24 June 2026
ACCEPTED 30 June 2026
PUBLISHED 24 July 2026

CITATION
Li Y, Luan M, Yu H and Yu H (2026) Improved Transformer-based detection of underwater plastic debris in complex environments.
Front. Mar. Sci. 13:1885615.
doi: 10.3389/fmars.2026.1885615

COPYRIGHT
© 2026 Li, Luan, Yu and Yu. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Improved Transformer-based detection of underwater plastic debris in complex environments

Yiyang Li[1], Mengsong Luan[1], Hanpeng Yu[1] and Huayun Yu[1,2]*

[1]School of Computer Science, Yangtze University, Jingzhou, Hubei, China, [2]Artificial Intelligence Research Platform, Yangtze University, Jingzhou, Hubei, China

Introduction: Underwater plastic debris detection remains challenging in visually cluttered environments because underwater images are often severely degraded, debris instances are small, and backgrounds are complex. To address these challenges, this study develops an improved RF-DETR detector for underwater plastic debris detection.

Methods: The proposed architecture integrates an underwater frequency-aware feature reweighting module, a multi-scale frequency-aware attention module, and a query adaptive reweighting module with bounded learnable scales. These modules are designed to enhance the representation of weak-texture and low-contrast features while improving query-level localization and classification. The model was evaluated on the TrashCan dataset and further assessed through a transfer experiment on the DeepTrash dataset.

Results: On the TrashCan dataset, the proposed detector achieved 93.11% precision, an F1-score of 86.63%, an mAP@50 of 91.84%, and an mAP@50:95 of 69.92%. In the DeepTrash transfer experiment, the TrashCan-trained checkpoint was adapted to the one-class plastic target dataset and achieved 91.67% precision, 79.00% recall, an F1-score of 84.86%, an mAP@50 of 86.23%, and an mAP@50:95 of 55.06%. Compared with representative baselines, including YOLOv11n, YOLO26n, YOLO26s, and RT-DETR, the improved RF-DETR achieved the strongest overall performance in the reported TrashCan evaluation and DeepTrash transfer experiment. Ablation results further showed that the three proposed modules performed best when used in combination.

Discussion: The results demonstrate that the proposed frequency-aware feature enhancement and adaptive query reweighting strategies improve the detection of small and low-contrast underwater debris. The transfer results also indicate promising cross-dataset generalization. Overall, the proposed method provides a favorable trade-off between detection performance and model complexity on the evaluated datasets.

KEYWORDS

complex underwater environments, frequency-aware feature reweighting, marine plastic debris, multi-scale refinement, query adaptive reweighting, Transformer-based detection, underwater object detection

1 Introduction

Marine plastic pollution has become a long-standing concern in marine conservation and ocean governance as plastic production and consumption continue to rise. Plastic waste is now documented throughout the water column, from near-surface layers to the seafloor, and its presence is associated with habitat damage, coral stress, and wider ecological disturbance. As a result, marine monitoring is increasingly built around sensor networks, automated interpretation, and robot-assisted surveys rather than manual inspection alone. Work on integrated observing systems, seafloor litter surveys, AI-based pollution monitoring, deep-sea autonomous observation, and underwater robotic autonomy reflects this operational change (Maximenko et al., 2019; Madricardo et al., 2020; Prakash and Zielinski, 2025; Barnard et al., 2024; Rohan et al., 2025).

Underwater plastic debris detection also differs markedly from litter detection on land because the visual evidence is less stable and the scene composition is far more variable. The target set spans recognizable objects such as bottles, cans, bags, and cups, but also films, fragments, ropes, and partly hidden residues whose outlines are incomplete or deformed. In this paper, two datasets are used, each serving a different purpose. The ablation experiments and part of the comparative experiments are conducted on the 22-class TrashCan dataset (Hong et al., 2020), while the transfer experiment is conducted on the one-class DeepTrash dataset (Tata et al., 2021). Survey papers repeatedly return to the same unresolved issues: impaired image quality, small targets, domain shift across scenes, and deployment constraints (Er et al., 2023; Jian et al., 2024). A detector intended for marine-debris monitoring therefore needs to remain reliable across both relatively clear views and scenes affected by clutter, low contrast, and changing acquisition conditions.

Performance typically deteriorates once attenuation, scattering, color cast, and non-uniform lighting erase fine contours and weaken object–background contrast. For that reason, many underwater detectors couple recognition with compensation for image degradation. MarineDetNet inserts image enhancement into the pipeline (Lakshmanan et al., 2025). Multi-distortion modeling likewise shows that degradation handling can change downstream detection quality (Liu et al., 2026), and methods combining enhancement with deformable convolution further blur the boundary between restoration and feature extraction (Guo et al., 2025). Other designs pursue the same goal through stronger feature extraction, fusion, and attention-based heads (Li et al., 2025a, b).

Many recent systems still favor YOLO-style formulations because they are easy to deploy and computationally efficient. LFN-YOLO, DyAqua-YOLO, and CEH-YOLO pursue that direction through lightweight reparameterization, dynamic adaptation, and composite enhancement (Liu et al., 2025; Li and Peng, 2025; Feng and Jin, 2024). For marine debris specifically, instance segmentation, UTNet, and improved YOLO variants show that streamlined architectures can still perform well in difficult scenes (Corrigan et al., 2023; Cui et al., 2025; Deluxni and Sudhakaran, 2025; Yuanwei et al., 2026). Yet the present task is not primarily defined by real-time operation or maximum frame rate. In deep-sea monitoring, where data transmission can be difficult and repeated acquisition is limited, the more important requirement is a reliable prediction from each available image rather than a decision accumulated through many rapid detections.

This priority becomes sharper in deep-sea inspection, fixed-point robotic grasping, and low-frequency sampling. These operating modes involve costly deployment, limited communication bandwidth, sparse opportunities for reacquisition, and little room for manual verification after the fact (Barnard et al., 2024; Rohan et al., 2025). When a target appears briefly or from a narrow viewpoint, useful progress means reducing misses and false alarms in each captured frame rather than only accelerating inference. Models that exploit broad scene context and allow task-specific decoder behavior are therefore appealing.

For this reason, using Transformer-based models is a promising choice. A series of existing studies has shown that DETR variants, such as UW-DETR and Efficient-Deformable-DETR, also achieve strong performance in underwater detection (Li et al., 2024; Xie et al., 2025; Lin et al., 2026). At the same time, a large body of research indicates that relying solely on local appearance is insufficient to effectively distinguish target objects from the surrounding background. In such cases, a global attention mechanism with a broader receptive field and stronger contextual connectivity can enable better performance in underwater detection, which is also the motivation for introducing the query adaptive reweighting module in this paper (Liu et al., 2022; Meng et al., 2021; Wang et al., 2022).

Among DETR-type models, RF-DETR is selected as the baseline for its searched feature pathway, end-to-end set prediction, and query-based global reasoning rather than for real-time inference as the core objective (Robinson et al., 2026). This design better matches the deep-sea monitoring setting emphasized in this paper: when data transmission is difficult and repeated acquisition is limited, each available image should support a reliable decision instead of relying on many rapid detections. Its real-time capability is therefore considered a secondary inherited advantage, while the main motivation is stable single-image prediction with a favorable balance between representation quality and model complexity. Compared with UWDETR, Efficient-Deformable-DETR, and RT-DETR, RF-DETR provides a more general searched detector scaffold for inserting the proposed frequency-domain, multi-scale, and query-side modifications. Such a scaffold is valuable for underwater plastic debris detection, where target size varies widely, morphology is unstable, and sources of background interference are diverse. At the same time, recent studies highlight the importance of frequency-aware representation in degraded visual scenes. Underwater frequency refinement, frequency channel attention, Fourier-domain modeling, global filtering, high-frequency multispectral attention, and multi-scale frequency fusion all suggest that frequency information can improve feature discrimination when texture is weak and background energy is dominant (Wang et al., 2025; Qin et al., 2021; Li et al., 2021; Rao et al., 2021; Ying et al., 2025; Yan et al., 2025; Liu et al., 2024). These observations motivate the integration of Fourier-domain band reweighting, multi-scale feature refinement, and query-level recalibration into the RF-DETR framework.

To clarify the baseline selection, Table 1 compares candidate detector families from the perspective of how the original detectors match underwater debris detection, rather than final detection scores

TABLE 1 Comparison of model characteristics for baseline selection.

Candidate detector	Detection formulation	Main advantage	Limitation for underwater debris	Reason for baseline suitability
Faster R-CNN	Two-stage region-based detector	Strong region proposal mechanism and stable localization for general objects	Heavier pipeline, proposal-dependent inference, and limited global relation modeling under dense small-object clutter	Useful classical reference, but less suitable as the main baseline because underwater debris detection requires stronger scene-level context and more compact end-to-end prediction
SSD	One-stage dense detector	Simple inference pipeline and moderate computational cost	Dense anchors depend strongly on local appearance and can miss weak-texture or low-contrast debris	Serves as a conventional efficiency-oriented baseline, but provides limited support for global context modeling
YOLO-series detectors	One-stage dense detectors using grid-, anchor-, or anchor-free assignment	High inference efficiency, mature lightweight implementations, and dense multi-scale prediction for visually clear debris categories	Predictions mainly depend on local feature cells and NMS, so weak boundaries, turbid backgrounds, and irregular debris morphology can increase missed detections, false positives, or duplicate boxes	Used as deployment-oriented comparison baselines, but not selected as the primary baseline because the local dense formulation is less aligned with the global-context and duplicate-suppression needs of cluttered underwater debris detection
RT-DETR	Real-time end-to-end Transformer detector with efficient hybrid encoding	Provides NMS-free prediction and Transformer decoding while maintaining practical real-time efficiency	Its design mainly emphasizes latency and efficient high-confidence query selection for general detection; in this study, speed is a secondary deployment advantage rather than the core goal	Used as a strong Transformer reference, but not selected as the main baseline because the task prioritizes reliable single-image prediction under limited data transmission and sparse reacquisition over repeated high-speed detection
RF-DETR	End-to-end Transformer detector with a searched feature pathway and query-based decoder	Combines global context modeling, NMS-free set prediction, query-level object reasoning, and an architecture-search driven accuracy–complexity balance	The original model is still general-purpose and does not explicitly compensate for underwater frequency attenuation, severe image degradation, or multi-scale debris ambiguity	Selected as the primary baseline because it better matches reliable single-image prediction in deep-sea monitoring: global context, set prediction, and query reasoning help reduce background confusion and duplicate detections for small or irregular debris, while real-time capability remains only an additional non-dominant advantage

alone. RF-DETR is selected as the baseline because the unmodified model already combines a searched feature pathway, end-to-end set prediction, NMS-free Transformer decoding, and global relationship modeling. These characteristics are useful for reliable single-image decisions in deep-sea debris monitoring: global context helps distinguish weak debris cues from visually similar backgrounds, set prediction reduces duplicate detections in clutter, query-level object reasoning is suitable for small or irregular targets, and the searched architecture provides a favorable accuracy–complexity trade-off when each transmitted frame is important. Therefore, RF-DETR is a structurally competitive detector for the baseline experiment itself. Its real-time capability is retained as an auxiliary deployment benefit, whereas the baseline selection is driven by reliability-oriented representation and decoding properties.

Based on this motivation, this paper develops Improved RF-DETR for underwater plastic debris detection. The model introduces an underwater frequency-aware feature reweighting module, a multi-scale frequency-aware attention module, and a query adaptive reweighting module, thereby strengthening feature modeling and query decoding under low-contrast, weak-texture, and cluttered conditions.

The main contributions of this paper are as follows:

1. We construct the Improved RF-DETR framework for underwater plastic debris detection. While preserving the end-to-end detection paradigm, the model performs coordinated optimization at the stages of feature modeling and query decoding.
2. We propose an underwater frequency-aware feature reweighting mechanism and a multi-scale frequency-aware attention mechanism. The former shifts the Fourier spectrum to the center, applies centered low-/high-frequency radial masks, and then reconstructs the feature by inverse shifting and inverse Fourier transform, whereas the latter fuses parallel multi-scale convolutions, forms a stable spatial low-/high-frequency approximation, and refines the encoder input through optional gating, dropout, and residual scaling.
3. We design a query adaptive reweighting strategy that learns a bounded scale for each base query, reuses the same base-query scales across grouped queries during training, and strengthens query response and matching stability for small, weak-texture, and easily confused targets during decoding.
4. We conduct systematic experiments on underwater debris datasets. Comparative experiments, ablation studies, and transfer validation support the effectiveness and practical relevance of the proposed method.

The remainder of this paper is organized as follows. Section 2 reviews common object detection models, as well as related work on underwater object detection and small-object detection. Section 3 introduces the architecture and improvements of the improved RF-DETR in detail. Section 4 describes the experimental configurations and settings. Section 5 focuses on the experimental results and analysis of observed phenomena. Section 6 concludes the paper and discusses future research directions.

2 Related work

2.1 General object detection frameworks

Recent underwater detectors increasingly borrow from DETR-style formulations instead of relying only on convolutional dense predictors. UW-DETR adapts RT-DETR with feature fusion for underwater imagery (Li et al., 2024); Efficient-Deformable-DETR uses deformable attention to keep computation manageable while improving small-target handling (Xie et al., 2025); and improved RT-DETR variants optimize feature fusion and feature extraction while reducing computational complexity and model size to some extent (Lin et al., 2026). What makes these models relevant here is the way global relational modeling compensates for unreliable local texture and irregular boundaries.

In real underwater scenes, many types of marine debris are occluded by other objects or mixed with surrounding materials. As a result, the model cannot reliably identify the correct targets based only on local regions. A large body of research has shown that incorporating surrounding background context into the decision process helps resolve ambiguous cues (Gao et al., 2024). For example, Liu's DAB-DETR combines queries with dynamic anchor boxes (Liu et al., 2022); Meng's Conditional DETR accelerates convergence by using conditional query constraints (Meng et al., 2021); and Wang's Anchor DETR optimizes localization performance through query initialization and spatial priors (Wang et al., 2022).

After comparing Transformer detector candidates, this paper adopts RF-DETR as the baseline model (Robinson et al., 2026). UW-DETR and improved RT-DETR are valuable underwater or efficient variants, but their original designs are already tied to specific fusion or efficiency strategies; Efficient-DeformableDETR mainly emphasizes deformable attention and computation control. RF-DETR instead offers a more general searched feature pathway with NMS-free set prediction and query reasoning, which is better suited to evaluating whether the proposed modules improve reliable single-image prediction under limited data transmission and sparse reacquisition. Frequency-oriented studies are also relevant: underwater frequency refinement (Wang et al., 2025) and frequency channel attention (Qin et al., 2021) both support redistributing spectral information when contrast is weak. That combination makes RF-DETR a natural base for adding frequency reweighting, multi-scale refinement, and query recalibration without abandoning end-to-end prediction.

2.2 Underwater object detection

Underwater object detection must contend with imaging degradation, scale variation, background interference, and deployment constraints. Recent reviews summarize these issues from the perspectives of image quality, small-object difficulty, dataset limitations, and practical deployment (Er et al., 2023; Jian et al., 2024).

A large body of work addresses degradation compensation before or during detection. MarineDetNet improves marine-object detection through image-quality enhancement (Lakshmanan et al., 2025). Multidistortion modeling likewise shows that explicit compensation for underwater degradation can materially influence downstream performance (Liu et al., 2026). Joint enhancement with deformable convolution further highlights the interaction between restoration and feature extraction (Guo et al., 2025). Other models strengthen target discrimination through enhanced feature extraction, fusion, and attention-based heads (Li et al., 2025a, b). Viewed collectively, this literature indicates that degradation handling is most effective when it is coupled with higher-level feature modeling rather than treated as a separate step.

In terms of detection backbones, the YOLO family remains the most widely used choice in underwater detection. LFN-YOLO improves small-object recognition through lightweight reparameterization and multi-scale fusion (Liu et al., 2025). DyAqua-YOLO enhances recognition in low-contrast and cluttered scenes through a dynamic adaptive architecture (Li and Peng, 2025). CEH-YOLO improves performance in complex underwater imagery through composite enhancement (Feng and Jin, 2024). Their continued use reflects clear engineering value for embedded platforms, online inspection, and applications that demand high frame rates.

For marine debris specifically, research has increasingly focused on irregular morphology, blurred boundaries, and confusion with the surrounding background. Corrigan et al. improve local contour description through instance segmentation (Corrigan et al., 2023). UTNet demonstrates the feasibility of real-time marine debris recognition from the standpoint of lightweight deployment (Cui et al., 2025). A YOLOv12-based debris detector and an improved YOLOv11 network reach similar conclusions: structural optimization can improve recognition in complicated underwater scenes while retaining the speed advantages of one-stage detectors (Deluxni and Sudhakaran, 2025; Yuanwei et al., 2026). Even so, existing methods still leave two gaps for complex underwater plastic debris detection. First, many of them prioritize lightweight design, real-time operation, and frame-rate advantages, whereas deep-sea monitoring often requires stable single-image prediction under limited data transmission and sparse reacquisition. Second, convolution-dominated one-stage detectors remain constrained by local receptive fields when backgrounds are large-scale, targets are irregular, and texture cues are weak. DETR-type models, especially RF-DETR, are therefore attractive alternatives when end-to-end set prediction and global relationship modeling are more critical (Robinson et al., 2026).

2.3 Small-object detection

Underwater plastic debris detection also inherits the challenges of small-object recognition. Targets such as plastic fragments, films, and ropes often occupy only a small number of pixels and are frequently accompanied by occlusion, deformation, and background interference. Under these conditions, edge, texture, and local structural cues are easily attenuated during deep semantic extraction. Prior work shows that preserving fine-grained information and coordinating features across scales are central to reliable small-object recognition. HTDet demonstrates the value of jointly modeling global semantics and local detail through a hybrid Transformer architecture (Chen et al., 2023). The autoencoding pyramid method further shows that cross-scale feature organization

and neighborhood information can improve small-target representation in complex underwater environments (Zhang et al., 2025).

Frequency-domain modeling provides an additional prior for small and weak-texture targets. FCANet reconstructs channel responses through frequency channel attention and shows that frequency information contributes positively to feature selection (Qin et al., 2021). Fourier Neural Operator highlights the expressive power of frequency space for global information representation (Li et al., 2021). Global Filter Networks emphasize the efficiency of global modeling through frequency filtering (Rao et al., 2021). In adjacent visual tasks, high-frequency multispectral attention has improved cross-domain generalization (Ying et al., 2025), while multi-scale frequency-enhanced Transformers and frequency-spatial dual-domain fusion networks show how frequency-aware refinement can preserve boundaries and details (Yan et al., 2025; Liu et al., 2024). These studies provide direct motivation for the underwater frequency-aware feature reweighting module and the multi-scale frequency-aware attention module used here.

Viewed together, prior work on small-object detection underscores the value of multi-scale representation, frequency-aware modeling, and attention-based refinement. For underwater plastic debris, however, small object difficulty is entangled with image degradation, background clutter, irregular morphology, and limited cross-scale semantics. Compared with detectors that rely mainly on local convolutional evidence or dense candidate screening, RF-DETR is structurally better aligned with this setting. Its end-to-end set-prediction mechanism reduces the loss of weak targets during candidate screening, and its global modeling helps disentangle small debris from cluttered backgrounds. RF-DETR is therefore both a reasonable baseline and a suitable substrate for the frequency reweighting, multi-scale refinement, and query recalibration introduced here.

3 Methods

This section presents the detector from the overall architecture down to the module-level changes that improve its underwater behavior. The discussion begins with Improved RF-DETR pipeline and then turns to the three added components: the underwater frequency-aware feature reweighting module (UFFR), the multi-scale frequency-aware attention module (MFA), and the query adaptive reweighting module (QAR). The following subsections trace these components in the order in which the network processes information—backbone feature reweighting, pre-encoder multi-scale refinement, and decoder-side query recalibration—so that their roles can be understood within the end-to-end structure of the baseline detector.

3.1 Overall framework

Building on RF-DETR (Robinson et al., 2026), this work constructs an enhanced end-to-end detector for underwater debris detection in complex scenes. The baseline Transformer detector is retained, while three targeted modifications are introduced at the stages of Fourier-domain feature reweighting, pre-encoder multi-scale enhancement, and query-adaptive reweighting. The resulting network includes a backbone feature extractor, a multi-scale feature construction and enhancement module, a Transformer encoder decoder, and a prediction head. The proposed additions are the underwater frequency-aware feature reweighting module (UFFR), the multi-scale frequency-aware attention module (MFA), and the query adaptive reweighting module (QAR).

When RF-DETR is transferred from general natural images to underwater debris detection, three limitations become prominent. First, the backbone features do not explicitly control the redistribution of low- and high-frequency information, so target-related detail can be overwhelmed by low-frequency background content. Second, the features delivered to the encoder lack a dedicated refinement stage for small targets and cluttered underwater scenes. Third, decoder queries enter attention with identical prior importance, even though their responsiveness to real targets can differ. UFFR, MFA, and QAR are introduced at these three stages: after the backbone, before the encoder, and before decoder attention, respectively. The overall architecture is shown in Figure 1.

Given an input image, the backbone first extracts deep visual features. UFFR then performs a two-dimensional FFT, shifts the spectrum so that low-frequency components are centered, applies centered radial low-/high-frequency reweighting, shifts the spectrum back, and reconstructs the feature by inverse FFT. The reweighted features are passed to the neck to build multi-scale representations. MFA is inserted between the neck output and the Transformer encoder. It applies parallel multi-scale convolutions, averages the branch responses, forms a stable spatial low-/high-frequency approximation through local averaging and residual decomposition, and finally refines the feature through optional gating, $\mathrm{Dropout}_{2d}$, and an optional residual connection with a learnable gain. In the decoding stage, QAR multiplies each base query by a bounded learnable scaling coefficient, applies the scales directly to a single query set during inference, and shares the same base-query scales across grouped queries during training.

With these additions, the detector preserves the end-to-end behavior of RF-DETR while introducing coordinated refinement in three places: Fourier-domain feature balancing, pre-encoder multi-scale enhancement, and query-side recalibration.

3.2 Underwater frequency-aware feature reweighting module

Convolutional backbone features are learned in the spatial domain and do not explicitly regulate the relative influence of different frequency bands. In underwater imagery, this is problematic because low-frequency background energy can dominate while detail-related frequencies associated with small targets are attenuated. Recent underwater studies suggest that frequency-level refinement can improve degraded feature representation (Wang et al., 2025), and FCANet likewise shows that frequency information can support discriminative feature selection (Qin et al., 2021). Motivated by these observations, the underwater frequency-aware feature reweighting module (UFFR) is inserted after the middle- and high-level backbone features. The corresponding mathematical

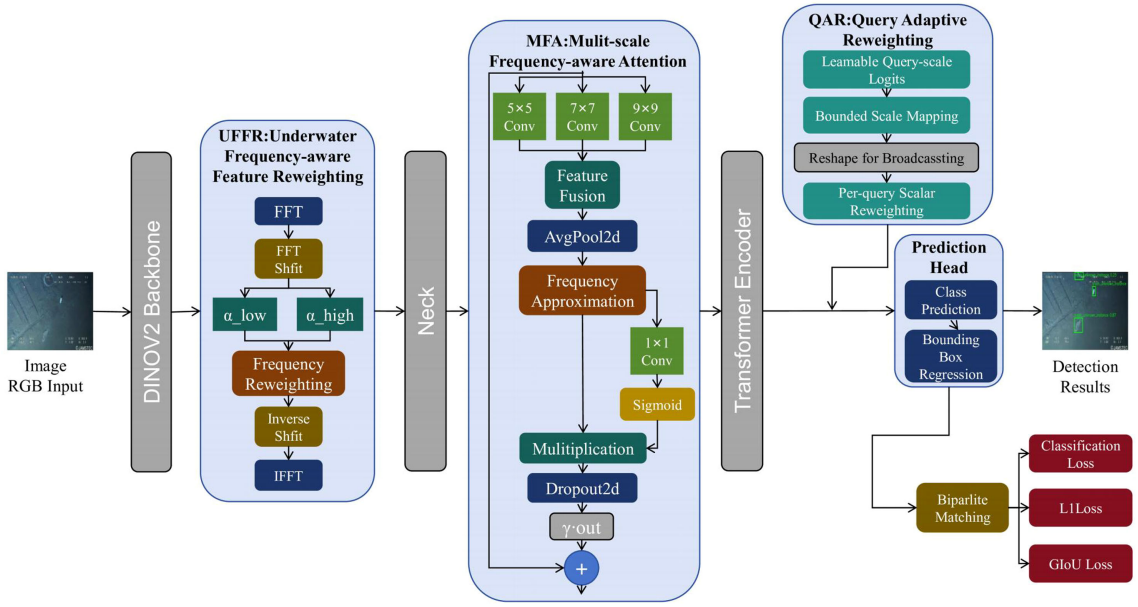


FIGURE 1
Overall architecture of the Improved RF-DETR model.

expressions are given in Equations 1–7. Its structure is shown in Figure 2.

Let the input feature map be:

$$\mathrm{F} \in \mathbb{R}^{B\times C\times H\times W} \tag{1}$$

where B, C, H, and W denote the batch size, number of channels, height, and width, respectively. The module expects a four-dimensional tensor with channel number C equal to the configured module width. In the implementation, the input is first cast to fp32 inside an autocast-disabled block, and a two-dimensional Fourier transform with orthonormal normalization is then applied along the last two dimensions:

$$\hat{\mathrm{F}} = \mathcal{F}_{\mathrm{ortho}}(\mathrm{F}) \in \mathbb{C}^{B\times C\times H\times W} \tag{2}$$

where $\mathcal{F}_{\mathrm{ortho}}(\cdot)$ denotes the two-dimensional FFT with norm="ortho". The spectrum is then shifted to the center by

$$\tilde{\mathrm{F}} = \mathcal{S}(\hat{\mathrm{F}}) \in \mathbb{C}^{B\times C\times H\times W} \tag{3}$$

where $\mathcal{S}(\cdot)$ denotes torch.fft.fftshift(·, dim=(-2,-1)), which moves low-frequency components to the center of the spatial-frequency plane.

To construct the centered radial masks, the implementation generates a normalized grid on the two spatial axes with torch.linspace (-1, 1, H) and torch.linspace(-1, 1, W), followed by a

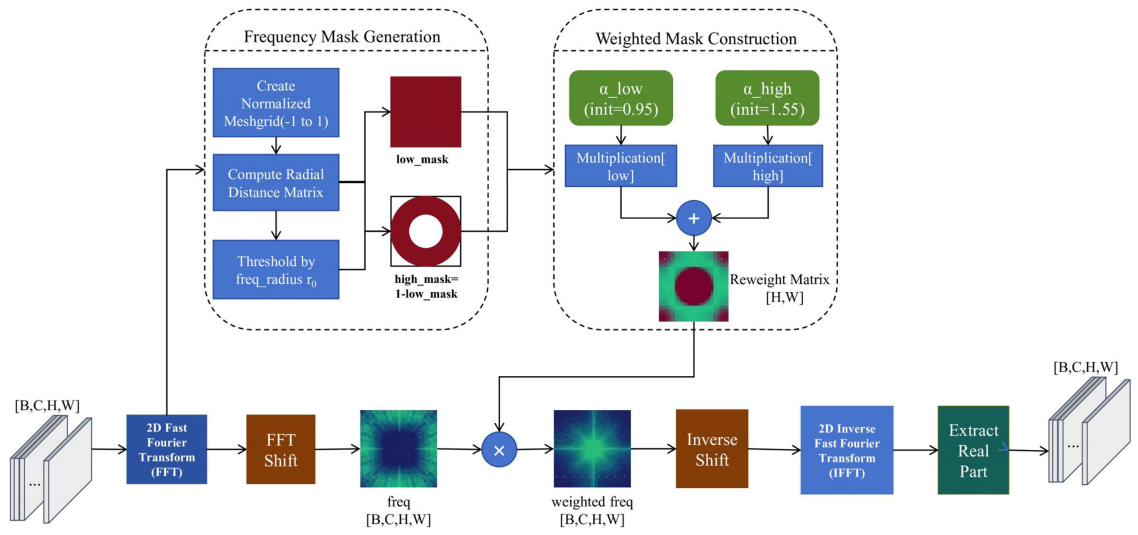


FIGURE 2
Architecture of the underwater frequency-aware feature reweighting module. The pipeline performs 2D FFT, FFT shift, centered low-/high-frequency mask reweighting, inverse FFT shift, and 2D IFFT reconstruction.

torch.meshgrid call with indexing="ij". The resulting coordinates are

$$u_h = -1 + \frac{2(h-1)}{H-1}, \qquad v_w = -1 + \frac{2(w-1)}{W-1}, \tag{4}$$
$$r(h,w) = \sqrt{u_h^2 + v_w^2}$$

where $h = 1,\ldots,H$ and $w = 1,\ldots,W$. Let r_0 denote freq_radius. In the reported experiments, $r_0 = 0.6$, and the corresponding masks are written as

$$\begin{cases} \text{low_mask}(h,w) = \text{II}(r(h,w) < r_0), \\ \text{high_mask}(h,w) = 1 - \text{low_mask}(h,w), \end{cases} \tag{5}$$

where $\text{low_mask}, \text{high_mask} \in \mathbb{R}^{H\times W}$ are two-dimensional masks. Here, low_mask corresponds to the centered circular region of the shifted spectrum, and high mask corresponds to the complementary outer region.

Two scalar coefficients, α_{low} and α_{high}, are then used to form the reweighting mask

$$M(h,w) = \alpha_{\text{low}}\ \text{low_mask}(h,w) + \alpha_{\text{high}}\ \text{high_mask}(h,w). \tag{6}$$

When use_learnable=True, α_{low} and α_{high} are implemented as learnable parameters; otherwise, they are fixed buffers. In the reported experiments, the learnable setting is used, with initialization values 0.95 and 1.55, respectively. The mask is broadcast across the batch and channel dimensions and multiplied element-wise with the shifted Fourier-domain feature. The weighted spectrum is then shifted back and reconstructed by inverse FFT, giving

$$F_r = \text{Re}\left(\mathcal{F}_{\text{ortho}}^{-1}\left(\mathcal{S}^{-1}\left(\tilde{F}\,\dot{o}\,M\right)\right)\right) \in \mathbb{R}^{B\times C\times H\times W} \tag{7}$$

where $\mathcal{S}^{-1}(\cdot)$ denotes torch.fft.ifftshift(·, dim=(-2,-1)), $\mathcal{F}_{\text{ortho}}^{-1}(\cdot)$ denotes the inverse transform with the same normalization, and $M \in \mathbb{R}^{H\times W}$ is broadcast to all B batches and C channels.

UFFR therefore applies a shared centered low-/high-frequency radial reweighting to the shifted Fourier-domain feature using freq_radius=0.6 and two scalar coefficients, followed by inverse shifting and inverse-transform reconstruction. In the present implementation, its effect comes from this centered band reweighting rather than from fine-grained frequency-selective attention.

3.3 Multi-scale frequency-aware attention

The encoder in RF-DETR performs global modeling, but its effectiveness still depends strongly on the quality of the features supplied by the neck. In underwater debris detection, small plastic fragments may closely resemble cluttered background patterns, and scale variation further complicates discrimination. MFA is therefore placed between the neck and the Transformer encoder to strengthen the encoder input through parallel multi-scale aggregation and a stable low-/high-frequency approximation implemented in the spatial domain. Related work has shown that multi-scale frequency-aware refinement can benefit boundary preservation and detail enhancement (Yan et al., 2025; Liu et al., 2024). The corresponding mathematical expressions are given in Equations 8–15. The module structure is shown in Figure 3.

In the baseline RF-DETR, the neck output is forwarded to the encoder after the required transformations, without an explicit refinement stage tailored to underwater small targets and clutter. MFA replaces this direct path by adding parallel convolutions, branch averaging, local low-pass/high-pass decomposition, an optional spatial gate, dropout regularization, and an optional residual connection.

Let the input feature be:

$$X \in \mathbb{R}^{B\times C\times H\times W} \tag{8}$$

where B, C, H, and W denote the batch size, number of channels, height, and width of the feature map, respectively. The module expects a four-dimensional tensor with channel number C equal to the configured module width.

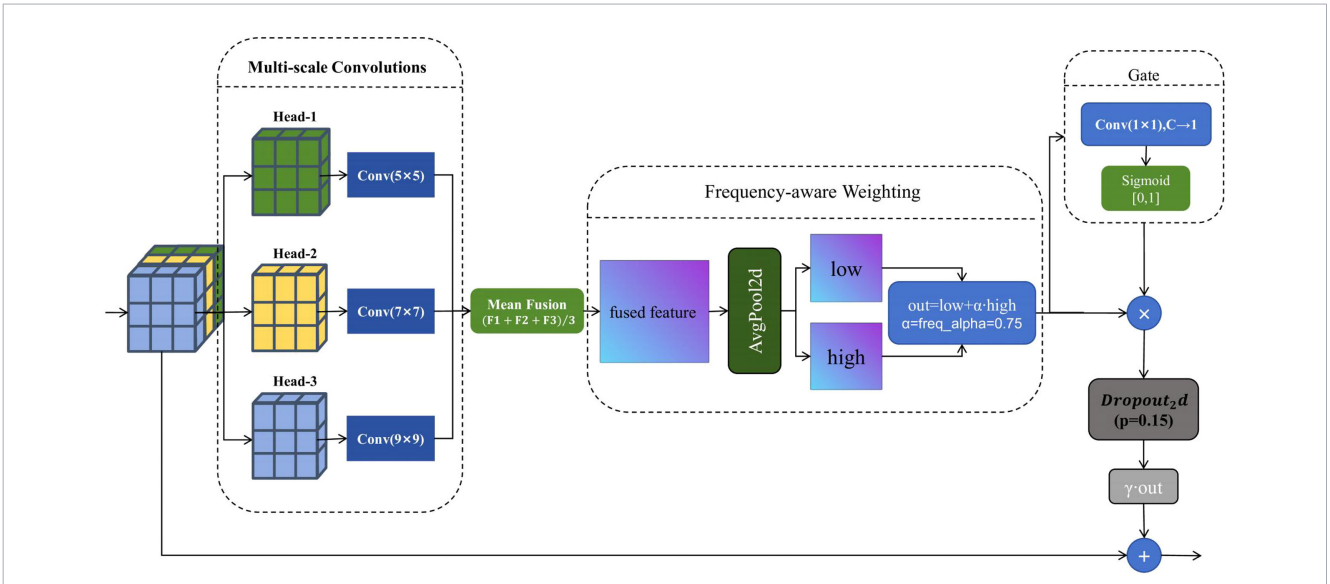


FIGURE 3
Architecture of the multi-scale frequency-aware attention module. Parallel multi-scale convolutions are averaged, decomposed into low- and high-frequency spatial components, and then refined by optional gating, dropout, and residual scaling.

Multiple convolution branches with different receptive fields produce a set of spatial representations. Let $\{\mathrm{F}^{(k)}\}_{k=1}^{K}$ denote the outputs of the K branches, where each $\mathrm{F}^{(k)} \in \mathbb{R}^{B\times C\times H\times W}$. In the reported implementation, $K = 3$ and the branch kernels are 5×5, 7×7, and 9×9, corresponding to mfascales=(1,2,3). The fused multi-scale feature is obtained by averaging the branch outputs:

$$\mathrm{X}_{\mathrm{ms}} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{F}^{(k)} \in \mathbb{R}^{B\times C\times H\times W} \tag{9}$$

The fused feature is then decomposed into a smoothed low-frequency term and a residual high-frequency term:

$$\mathrm{X}_{\mathrm{low}} = \underset{3\times 3}{\mathrm{AvgPool}}(\mathrm{X}_{\mathrm{ms}}) \in \mathbb{R}^{B\times C\times H\times W} \tag{10}$$

$$\mathrm{X}_{\mathrm{high}} = \mathrm{X}_{\mathrm{ms}} - \mathrm{X}_{\mathrm{low}} \tag{11}$$

A scalar frequency-balancing coefficient $\alpha_{\mathrm{mfa}} \in \mathbb{R}$ scales the high-frequency residual:

$$\mathrm{X}_{\mathrm{freq}} = \mathrm{X}_{\mathrm{low}} + \alpha_{\mathrm{mfa}}\ \mathrm{X}_{\mathrm{high}} \in \mathbb{R}^{B\times C\times H\times W} \tag{12}$$

In the reported experiments, $\alpha_{\mathrm{mfa}} = 0.75$. This operation provides a stable frequency-aware approximation in the spatial domain rather than an explicit Fourier-domain spectrum scaling.

When the gate branch is enabled, a 1×1 convolution generates a single-channel gate from $\mathrm{X}_{\mathrm{freq}}$:

$$\mathrm{G} = \sigma(Conv_{1\times 1}(X_{\mathrm{freq}})) \in [0,1]^{B\times 1\times H\times W} \tag{13}$$

$$X_{\mathrm{gate}} = \begin{cases} X_{\mathrm{freq}} \odot \mathrm{G}, & \text{if the gate branch is enabled,} \\ X_{\mathrm{freq}}, & \text{otherwise,} \end{cases} \tag{14}$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function, and $\odot$ denotes element-wise multiplication. Because G has a single channel, it is broadcast along the channel dimension when multiplied with $\mathrm{X}_{\mathrm{freq}}$.

Finally, $\mathrm{Dropout}_{2\mathrm{d}}$ regularizes the gated feature, and an optional residual connection with learnable scalar gain γ produces the output:

$$X_{\mathrm{out}} = \begin{cases} X + \gamma\ \ \mathrm{Dropout}_{2\mathrm{d}}(\mathrm{X}_{\mathrm{gate}}), & \\ & \text{if the residual branch is enabled,} \\ \mathrm{Dropout}_{2\mathrm{d}}(\mathrm{X}_{\mathrm{gate}}), & \text{otherwise,} \end{cases} \tag{15}$$

In the reported experiments, both the gate and residual branches are enabled, the dropout rate is 0.15, and γ is initialized to 10^{-3}. A final torch.nan_to_num operation is applied in the implementation for numerical stabilization. The optional visualization branch only stores intermediate tensors and does not alter the forward computation.

As implemented here, MFA derives its effect from the combination of multi-scale spatial aggregation, local low-/high-frequency decomposition, optional gated spatial refinement, dropout regularization, and residual scaling. It is therefore a frequency-aware spatial approximation rather than a Fourier-domain attention block.

3.4 Query adaptive reweighting module

In standard DETR-style decoding, a fixed set of queries is processed with equal prior importance. That assumption can be restrictive in underwater scenes, where only some queries may respond well to small, dense, or low-contrast targets. The query adaptive reweighting module (QAR) addresses this issue by learning a bounded scale for each base query index before attention is computed. The design is related to prior work on query priors and conditional query construction (Liu et al., 2022; Meng et al., 2021; Wang et al., 2022). The corresponding mathematical expressions are given in Equations 16–19. Its structure is shown in Figure 4.

The original RF-DETR uses a fixed number of queries and does not explicitly distinguish their relative importance before decoder attention. QAR is inserted between the decoder input queries and the subsequent attention computation, where it rescales the original query tensor through query-wise coefficients. In the current implementation, these coefficients are learned parameters shared across samples rather than per-image dynamic predictions.

Let the input query tensor of the Transformer decoder be:

$$Q \in \mathbb{R}^{B\times N\times d} \tag{16}$$

where B denotes the batch size, N denotes the number of query vectors, and d denotes the feature dimension.

Let $N_q$denote the number of base queries, and let G denote group_detr. The module maintains a learnable logit vector $\in \mathbb{R}^{Ng}$ whose bounded query-wise scales are written as

$$\mathrm{s} \in \mathbb{R}^{N_q},\ s_i = s_{\min} + (s_{\max} - s_{\min})\ \sigma(\ell_i), \quad i = 1,2,\ldots,N_q \tag{17}$$

where $\sigma(\cdot)$ denotes the Sigmoid function. In the reported experiments, $s_{\min} = 0.5$, $s_{\max} = 2$, and $s_{\mathrm{init}} = 1$. The learnable logits are initialized by applying the inverse-Sigmoid mapping to the normalized initial scale after clipping it to $(10^{-4}, 1 - 10^{-4})$.

During inference or evaluation, when $N = N_q$, $\mathbf{s}$ is reshaped to $\mathbb{R}^{1\times N_q\times 1}$ and broadcast across the batch and feature dimensions. The resulting scaled query tensor is

$$Q'_{b,i,j} = s_i\ Q_{b,i,j},\ b = 1,\ldots,B,\ \ i = 1,\ldots,N_q,\ \ j = 1,\ldots,d \tag{18}$$

During training with grouped queries, when $N = N_qG$, the tensor is first reshaped to $\tilde{Q} \in \mathbb{R}^{B\times G\times N_q\times d}$, and the same base-query scales are shared across all G groups:

$$\tilde{Q}'_{b,g,i,j} = s_i\ \ \tilde{Q}_{b,g,i,j},\ b = 1,\ldots,B,\ g = 1,\ldots,G,\ \ i = 1,\ldots,N_q,\ \ j = 1,\ldots,d\,. \tag{19}$$

After grouped scaling, the tensor is reshaped back to $\mathbb{R}^{B\times N\times d}$. If N is neither $N_q$nor N_qG, the implementation raises a shape-mismatch error. A final torch.nan_to_num operation is applied for numerical stabilization.

QAR therefore introduces differentiated query modulation instead of treating all queries as equally informative. In the present implementation, this effect comes from bounded learnable base-query priors shared across samples, reused directly for single-group

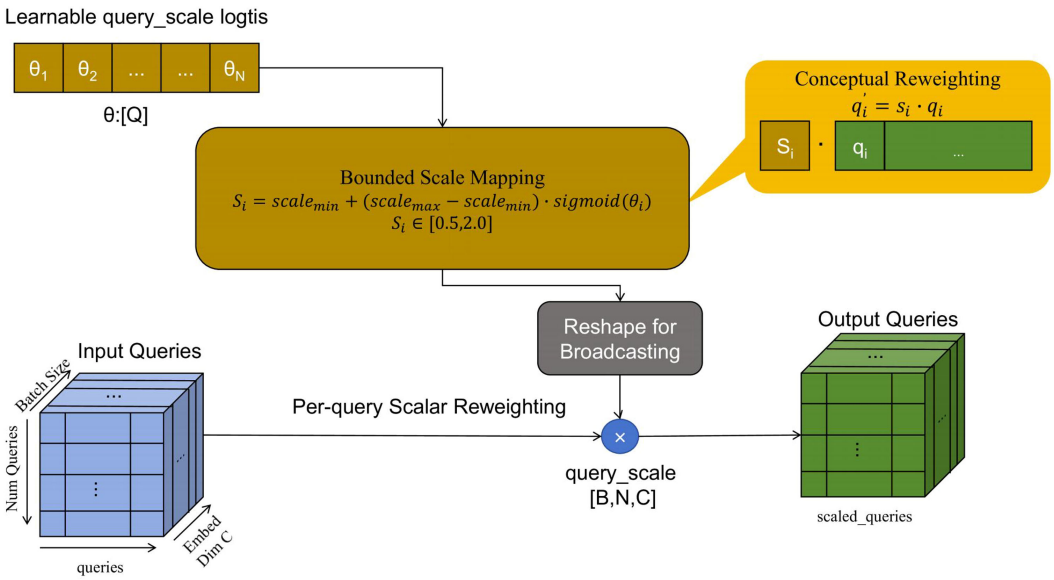


FIGURE 4
Architecture of the query adaptive reweighting module. Base-query scales are generated from bounded learnable logits, applied directly to a single query set during inference, and shared across grouped queries during training.

inference, and shared across grouped queries during training, which can improve matching stability and reduce interference in classification and localization, particularly for small or difficult targets.

4 Experimental setup

This section details the experimental protocol used to evaluate the proposed detector. It first introduces the datasets and data splits used for in-domain training and transfer fine-tuning, then defines the evaluation metrics, and finally describes the computational environment and training configuration. The aim is to keep the reported results transparent and reproducible.

4.1 Datasets

This paper uses two underwater debris datasets. TrashCan-Instance serves as the source dataset for model training, ablation experiments, and the main comparative evaluation (Hong et al., 2020), whereas DeepTrash (Tata et al., 2021) is used as the target dataset for transfer fine-tuning and testing.

TrashCan-Instance contains 7,212 annotated images spanning 22 categories. For this work, the dataset is randomly split into training, validation, and test subsets at 70%:20%:10%, yielding 5,048, 1,442, and 722 images, respectively. The random seed is fixed at 42. As a public underwater debris benchmark, TrashCan mainly covers three broad groups of targets: underwater debris, remotely operated vehicles, and benthic fauna and flora (Hong et al., 2020). Most images originate from the J-EDI dataset, and were extracted from videos collected over many years in the Sea of Japan by JAMSTEC-operated ROVs. The class-instance distribution of TrashCan is shown in Figure 5.

The DeepTrash dataset contains 2,866 images in total, including 2,151 training images and 715 test images (Tata et al., 2021). It contains only one target class, Plastic. The images are selected from aquatic plastic-debris field videos captured at different California locations, including South Lake Tahoe, Bodega Bay, and San Francisco Bay. DeepTrash does not overlap with TrashCan in image content, and its one-class label space differs from the 22-category TrashCan setting.

In this paper, the ablation experiments and part of the comparative experiments are conducted on TrashCan, while the transfer experiment is conducted on DeepTrash. In the transfer experiment, the proposed model keeps the hyperparameter configuration reported in Subsection 4.4 unchanged when moving from TrashCan to DeepTrash, and the weights trained on the 22-class TrashCan dataset are loaded first. Because DeepTrash contains only the Plastic class, the category prediction head is replaced with a one-class head and reinitialized before target-domain fine-tuning. The remaining comparison models follow the same configuration policy used in the comparative evaluation: their official recommended default settings are retained and the training epoch number is unified to 50.

4.2 Evaluation metrics

In this paper, Precision (P), Recall (R), F1-score, mean Average Precision (mAP), giga floating-point operations (GFLOPs), and the number of parameters are used to evaluate model effectiveness. Precision reflects the accuracy of positive predictions, whereas Recall measures the ability to identify positive samples. The F1-score is the weighted harmonic mean of Precision and Recall, and mAP represents the mean Average Precision across all categories. GFLOPs is a commonly used indicator for measuring computational complexity, and the number of parameters indicates model size. These metrics are widely used for evaluating object detection tasks, and their formulas are given in Equations 20–24.

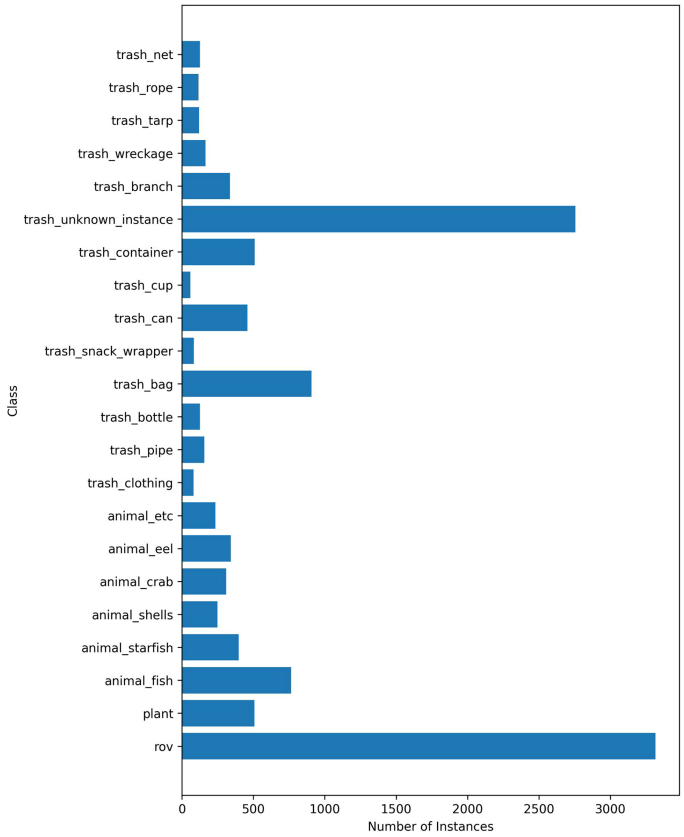


FIGURE 5
Category distribution of instances in the TrashCan dataset. Source: Based on data from Hong et al. (2020).

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{20}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{21}$$

$$\mathrm{F1} = 2 \cdot \frac{\mathrm{P} \cdot \mathrm{R}}{\mathrm{P} + \mathrm{R}} \tag{22}$$

$$\mathrm{AP} = \begin{cases} \displaystyle\int_0^1 P(r) \quad \mathrm{d}r, & \text{continuous PR curve} \\ \displaystyle\sum_{n=1}^{N} P_n \quad \Delta R_n, & \text{discrete PR curve} \end{cases} \tag{23}$$

$$\mathrm{mAP} = \frac{1}{K} \sum_{i=1}^{K} \mathrm{AP}(i) \tag{24}$$

Here, TP, FP, TN, and FN represent the numbers of true positives, false positives, true negatives, and false negatives, respectively. $P(r)$ represents the precision–recall curve, and K denotes the number of classes in the current recognition task. In object detection tasks, mAP depends on the selected Intersection over Union (IoU) threshold. In particular, mmAP@50 refers to the mean Average Precision achieved when the IoU threshold is set to 0.5.

4.3 Experimental parameters

All experiments were conducted on an ASUS TUF Gaming 5 Pro laptop running Windows 11. The system was equipped with an Intel Core i9-14900HX processor at 2.2 GHz, 48 GB of system memory, and an NVIDIA GeForce RTX 4060 Laptop GPU with 8 GB of video memory. Training was implemented in PyTorch 2.5.1 +cu121, and the CUDA driver version was 12.9. The detector was built on the DETR-based PyTorch implementation released by Roboflow. To reduce fluctuations caused by random initialization and data partitioning, the random seed was fixed at 42 in all formal experiments except during the Optuna search stage.

4.4 Training strategy and hyperparameter settings

Training uses the AdamW optimizer together with a learning-rate schedule that combines warmup and cosine annealing. During the initial iterations, the learning rate is gradually increased from a small value to the preset initial value in order to stabilize early optimization. It is then progressively decayed by cosine annealing. Automatic mixed precision (AMP) is enabled to reduce GPU memory usage and improve training efficiency, and gradient clipping is applied to suppress abnormal updates.

Rather than relying only on manually selected settings, this paper uses the Optuna framework with Bayesian optimization to search for suitable hyperparameters automatically. The search is performed on TrashCan with the TPE sampler and the median pruner for 50 trials, using validation mAP@50 as the optimization target. The search space includes input resolution, batch size, base feature dimension, learning rate, weight decay, dropout rate, gradient-clipping threshold, and the number of training epochs. The final configuration is input resolution 384×384, batch size 8, base feature dimension 12, learning rate 8.02×10^{-5}, weight decay 1.63×10^{-5}, dropout rate 0.25, gradient-clipping threshold 0.19, and a maximum training epoch number of 50. The 384×384 input resolution is used because it is the resolution selected by the hyperparameter search under the validation mAP@50 objective and because it provides a practical balance for the available 8 GB GPU. Compared with smaller inputs, it preserves more spatial detail for small and low-contrast debris after resizing; compared with larger inputs, it avoids excessive memory consumption, unstable batch sizing, and a disproportionate increase in training and inference cost. Therefore, this resolution is treated as a compromise among small-target detail preservation, detection accuracy, GPU memory usage, and computational efficiency rather than as an arbitrary setting. Early stopping is also enabled. When the monitored metric fails to improve by at least 0.01 for several consecutive epochs, training is terminated and the parameters corresponding to the best validation mAP@50 are retained. Unless otherwise stated, experiments involving the proposed model and its ablation variants use this configuration.

For the network and loss configuration, the backbone is set to dinov2_windowed_small. The number of ViT backbone layers is 12, and the number of decoder layers is 2. Both num_queries and num_select are set to 300, and group_detr is set to 13. The loss function jointly optimizes classification loss, bounding-box regression loss, and GIoU loss, where cls_loss_coef, bbox_loss_coef, and giou_loss_coef are set to 1, 5, and 2, respectively, while auxiliary loss is retained. In addition, ia_bce_loss is enabled, varifocalloss is disabled, and focal_alpha is set to 0.25. In Hungarian matching, set_cost_class = 2, set_cost_bbox = 5, and set_cost_giou = 2. For the added modules, UFFR uses freq_radius=0.6 and use_learnable=True, with alpha_low and alpha_high initialized to 0.95 and 1.55, respectively. In the forward pass, UFFR applies fftshift before centered mask reweighting and ifftshift before inverse FFT. MFA uses mfascales=(1,2,3), mfa_freq_alpha=0.75, mfa_dropout=0.15, gate=True, residual=True, enable_vis=False, and residual gain initialization init_scale=10^{-3}. QAR uses query_scale_init=1, scale_min=0.5, scale_max=2, scale_lr_factor=1, and group_detr=13; during training, the same base-query scales are shared across the 13 grouped query sets, whereas inference uses a single query group with $N = 300$. These settings remain aligned with the RF-DETR training paradigm while better matching the present task, which requires both classification stability and accurate localization.

5 Results

5.1 Ablation study

To verify the detection performance of the proposed model and explore the influence of different modules, ablation experiments were conducted based on RF-DETR. The results are shown in Table 2, where the best results are highlighted in bold. Group 1 evaluates the original RF-DETR baseline, and Groups 2–8 introduce UFFR, MFA, and QAR either individually or in combination. From the baseline model to the final Improved RF-DETR model, all ablation groups use the same hyperparameter configuration, and the specific numerical values are reported in Subsection 4.4. Only the module composition is changed across groups, ensuring that the differences in Table 2 reflect the effects of UFFR, MFA, and QAR rather than hyperparameter variation.

The original model achieves 83.59% Precision, 70% Recall, 76.2% F1, 82.98% mAP@50, and 60.53% mAP@50:95. Introducing a single module improves the baseline in different ways. UFFR raises Recall to 77% and mAP@50 to 88.05%, indicating that frequency-aware reweighting helps recover weak debris cues. MFA obtains the highest single-module Precision of 88.28% and improves mAP@50:95 to 67.26%, showing stronger feature discrimination and localization quality. QAR increases Recall to 81% and F1 to 82.33%, which suggests that query-level reweighting is especially useful for reducing missed detections.

The combined-module results further clarify the different roles of the three modules and their interaction. UFFR mainly

TABLE 2 Ablation study on the TrashCan dataset.

Group	UFFR	MFA	QAR	P	R	F1	mAP@50	mAP@50:9	5 FPS
1				83.59	70	76.2	82.98	60.53	**51.25**
2	✓			84.85	77	80.74	88.05	66.24	50.03
3		✓		88.28	76	81.68	88.58	67.26	43.76
4			✓	83.71	81	82.33	88.68	66.91	41.29
5		✓	✓	89.59	81	85.08	90.08	**70.39**	45.19
6	✓	✓		83.04	**85**	84.01	89.64	69.29	42.33
7	✓		✓	87.04	**85**	86.01	91.32	68.54	50.82
8	✓	✓	✓	**93.11**	81	**86.63**	**91.84**	69.92	49.86

✓ indicates the introduced module, and the best results are highlighted in bold. Source: Based on data from Hong et al. (2020).

strengthens degraded frequency cues: used alone, it raises Recall from 70% to 77% and mAP@50 from 82.98% to 88.05%, and when paired with MFA or QAR it helps both two-module variants reach the highest Recall of 85%. MFA contributes more to feature discrimination and localization quality: used alone, it gives the highest single-module Precision of 88.28% and mAP@50:95 of 67.26%, and the MFA+QAR combination further obtains the overall best mAP@50:95 of 70.39%. QAR mainly improves query matching and missed-target recovery: used alone, it raises Recall to 81% and F1 to 82.33%, and its combination with UFFR reaches 91.32% mAP@50. When all three modules are used together, the model does not maximize every single metric, but it achieves the strongest overall metric performance, with the best Precision (93.11%), F1-score (86.63%), and mAP@50 (91.84%) while maintaining 49.86 FPS. Compared with the original RF-DETR baseline, Improved RF-DETR improves Precision, Recall, F1, mAP@50, and mAP@50:95 by 9.52, 11, 10.43, 8.86, and 9.39 percentage points, respectively, showing that frequency reweighting, multi-scale refinement, and query recalibration act complementarily rather than simply adding redundant capacity.

5.2 Comparative evaluation

Improved RF-DETR is compared with Faster R-CNN, SSD, YOLOv8n, YOLOv8s, YOLOv11n, YOLOv11s, YOLO26n, YOLO26s, and RT-DETR on TrashCan. Table 3 contains the numerical comparison, and Figure 6 places those results against model size and computation. The inclusion of YOLOv8s and YOLOv11s makes the comparison cover both nano-scale and small-scale YOLO variants. To ensure consistency and reproducibility, all comparison models except the proposed model are trained with their official recommended default parameter configurations, and their training epoch number is uniformly set to 50. The proposed model uses the final hyperparameters obtained by Optuna because the three task-specific modules are strongly coupled with its network structure and training behavior. Directly transferring these optimized values to other detectors could introduce structure-related bias. Therefore, the other detectors are not re-tuned with the proposed model's optimal hyperparameters; instead, their official default configurations are retained for a relatively consistent and reproducible comparison.

On TrashCan, Improved RF-DETR records the highest Precision (93.11%), F1-score (86.63%), mAP@50 (91.84%), and mAP@50:95 (69.92%) in the comparison set. Faster R-CNN obtains the highest Recall of 84.52%, but the proposed model remains only 3.52 percentage points lower in Recall while surpassing Faster R-CNN by 6.21 points in Precision, 0.93 points in F1, 2.96 points in mAP@50, and 0.92 points in mAP@50:95. This indicates that the proposed detector favors a stronger overall detection balance rather than maximizing Recall alone.

Among the YOLO baselines, the newly added YOLOv8s is the strongest YOLO variant on F1, mAP@50, and mAP@50:95, reaching 85.07%, 88.19%, and 68.53%, respectively. YOLO26s provides the highest YOLO Recall of 84.02%, whereas YOLOv8n gives the highest YOLO Precision of 88.28%. Even against YOLOv8s, Improved RF-DETR improves Precision, F1, mAP@50, and mAP@50:95 by 5.83, 1.56, 3.65, and 1.39 percentage points, respectively. The Recall of YOLOv8s is 1.97 points higher, but its overall metric performance remains lower than that of the proposed detector.

Figure 6 makes the accuracy–complexity trade-off clearer. YOLO26n, YOLOv11n, and YOLOv8n occupy the extremely lightweight region, whereas YOLO26s, YOLOv11s, and YOLOv8s fill the middle range between nano models and larger detectors. Compared with YOLOv11s, Improved RF-DETR uses a similar amount of computation (21.98 versus 21.7 GFLOPs) but improves F1 and mAP@50 by 2.21 and 3.94 points. Compared with YOLOv8s, the proposed method uses fewer GFLOPs (21.98 versus 28.7) while still obtaining higher F1 and mAP@50. Relative to RT-DETR, it reduces computation from 109 GFLOPs to 21.98 GFLOPs while improving F1 from 77.9% to 86.63% and mAP@50 from 84.65% to 91.84%. Therefore, the proposed detector occupies a favorable region in Figure 6: it is not the smallest model, but it provides the best accuracy among the evaluated detectors at moderate computational cost.

TABLE 3 Comparison of detector performance on the TrashCan dataset.

Dataset	Model	Precision	Recall	F1	mAP@50	mAP@50:95	Parameters/M	GFLOP s
TrashCan Dataset	Faster R-CNN	86.9	84.52	85.7	88.88	69	41.4	135
	SSD	86.55	84	85.26	87.37	68.4	24.31	116.2
	YOLOv8n	88.28	80.9	84.43	87.08	66.09	3.01	4.05
	YOLOv8s	87.28	82.97	85.07	88.19	68.53	11.13	28.7
	YOLOv11n	85.93	82.63	84.25	87.51	66.91	2.58	3.1
	YOLOv11s	86	82.9	84.42	87.9	66.9	9.8	21.7
	YOLO26n	82.82	75.16	78.81	82.51	62.58	**2.37**	**2.6**
	YOLO26s	85.05	84.02	84.53	87.88	67.78	9.47	10.28
	RT-DETR	84.86	72	77.9	84.65	61.57	32.8	109
	OURS	**93.11**	**81**	**86.63**	**91.84**	**69.92**	28.87	21.98

The best results are highlighted in bold.

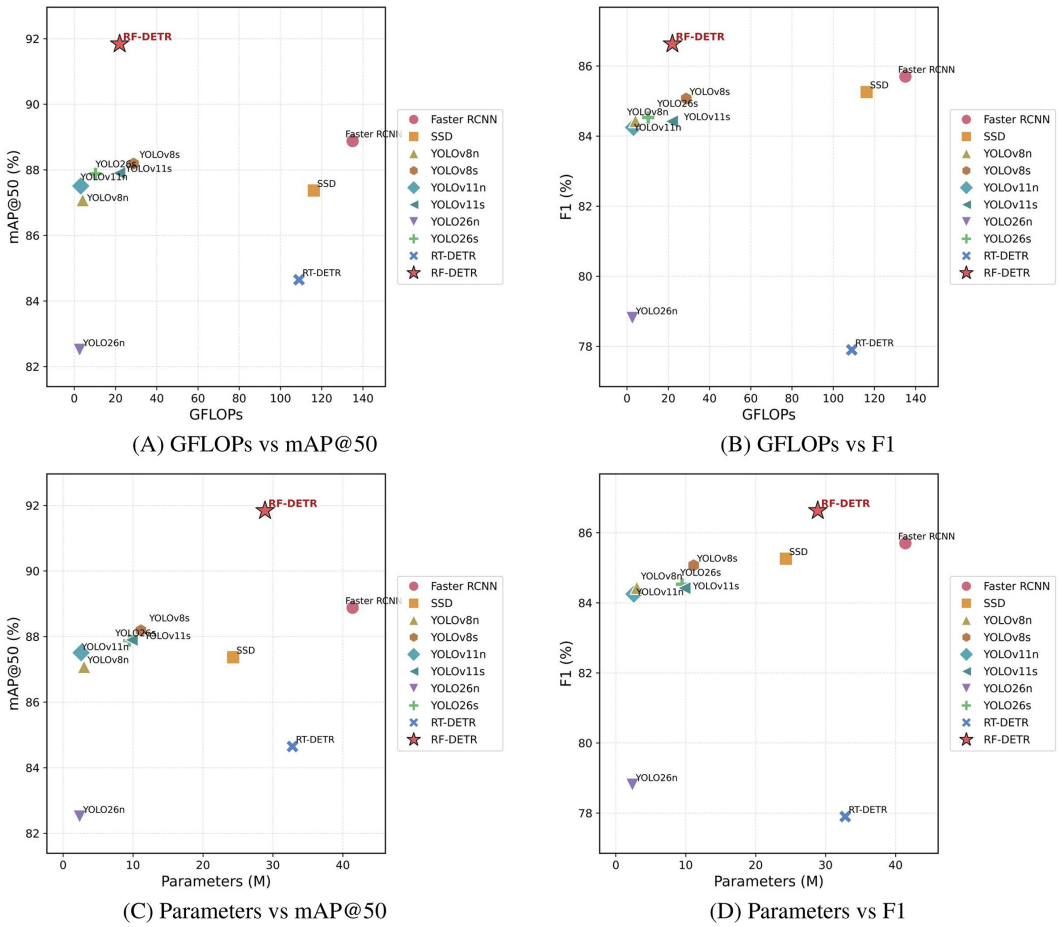


FIGURE 6
Complexity–performance comparison of different detectors on the TrashCan dataset. **(A–D)** show the relationships of GFLOPs with mAP@50, GFLOPs with F1, Parameters with mAP@50, and Parameters with F1, respectively. Source: Based on data from Hong et al. (2020).

5.3 Transfer experiment

The transfer experiment uses TrashCan as the source dataset and DeepTrash as the target dataset. For Improved RF-DETR, the hyperparameter configuration in Subsection 4.4 is kept unchanged during the transfer from TrashCan to DeepTrash, and the checkpoint trained on TrashCan is loaded before target-domain adaptation. Since DeepTrash contains only the Plastic class, the original 22-class category head is replaced with a one-class head and reinitialized before fine-tuning on the 2,151-image DeepTrash training split and evaluation on the 715-image test split. For the remaining detectors, the same policy as the comparative evaluation is used: their official recommended default configurations are retained and the training epoch number is set to 50. Therefore, Table 4 reports transfer fine-tuning under unchanged proposed-model hyperparameters and official-default comparison settings rather than direct zero-shot inference on DeepTrash.

After transfer fine-tuning on DeepTrash, Improved RF-DETR reaches 91.67% Precision, 79% Recall, 84.86% F1, 86.23% mAP@50,

TABLE 4 **Transfer experiment results on the DeepTrash dataset.**

Dataset	Model	Precision	Recall	F1	mAP@50	mAP@50:95	Parameters/M	GFLOP s
DeepTrash Dataset	YOLOv8n	84.67	73.7	78.81	81.93	49.54	3.01	4.04
	YOLOv8s	89.02	75.72	81.84	85.27	52.46	11.13	14.22
	YOLOv11n	86.98	74.37	80.18	84.08	51.36	2.58	3.16
	YOLOv11s	85.73	76.01	80.58	83.25	49.99	9.41	10.65
	YOLO26n	88.85	78.02	83.08	85.05	53.65	**2.38**	**2.6**
	YOLO26s	90.59	74.86	81.98	84.97	53.09	9.47	10.26
	RT-DETR	91.48	73	81.2	81.83	52.16	36.4	89.1
	OURS	**91.67**	**79**	**84.86**	**86.23**	**55.06**	28.82	31.35

The best results are highlighted in bold. Source: Based on data from Hong et al. (2020).

and 55.06% mAP@50:95. It ranks first in all five detection metrics among the compared detectors. Relative to YOLOv11n, Improved RF-DETR improves Precision, Recall, F1, mAP@50, and mAP@50:95 by 4.69, 4.63, 4.68, 2.15, and 3.7 points, respectively. Relative to the updated RT-DETR results, Improved RF-DETR improves the same five metrics by 0.19, 6, 3.66, 4.4, and 2.9 points, respectively, while using 7.58 M fewer parameters and reducing computation by 57.75 GFLOPs.

The YOLO variants show different transfer behaviors. YOLO26n remains the most efficient detector, with 2.38 M parameters and 2.6 GFLOPs, and it obtains the second-highest mAP@50:95 of 53.65%. YOLOv8s achieves the highest YOLO mAP@50 of 85.27%, while YOLO26s gives the highest YOLO Precision of 90.59%. Even so, Improved RF-DETR remains higher than all YOLO variants on Precision, Recall, F1, mAP@50, and mAP@50:95 after transfer fine-tuning.

The updated RT-DETR row changes the interpretation of the mAP@50:95 metric. Improved RF-DETR is now highest on mAP@50:95, followed by YOLO26n at 53.65%, YOLO26s at 53.09%, YOLOv8s at 52.46%, and RT-DETR at 52.16%. Under this transfer setup, Improved RF-DETR therefore provides the strongest overall metric performance on the one-class DeepTrash target dataset.

These results should be interpreted as a transfer experiment. The detector reuses representations learned from TrashCan and then adapts a reinitialized one-class head to the DeepTrash Plastic label. The comparison reflects performance after this training procedure, not direct use of a source-trained model without DeepTrash fine-tuning.

5.4 Qualitative results and error analysis

Figure 7 contrasts predictions from the baseline RF-DETR and Improved RF-DETR on representative difficult samples. Green, red,



FIGURE 7
Comparison of detection results between the baseline RF-DETR and the Improved RF-DETR **(A–H)**. Green, red, blue, and yellow boxes denote true positives, false positives, false negatives, and duplicate detections, respectively. Source: Based on data from Hong et al. (2020).

blue, and yellow boxes denote true positives (TP), false positives (FP), false negatives (FN), and duplicate detections, respectively.

These box labels are obtained by matching model predictions with manually annotated ground-truth boxes. A predicted box is counted as a TP and marked in green only when its IoU with the corresponding ground-truth box is higher than the preset threshold and the category labels are consistent. A prediction that cannot be matched to any ground-truth object in both spatial overlap and category label is counted as an FP and marked in red. If a ground-truth object has no matched prediction satisfying both the IoU threshold and category-consistency requirement, the ground-truth annotation box is marked as an FN in blue, indicating a missed target. For the same ground-truth object, when multiple predicted boxes satisfy the matching conditions, only the prediction with the highest confidence or best matching quality is retained as the TP; the remaining predictions that repeatedly match the same object are marked as duplicate detections in yellow.

When only a few objects are visible, the baseline still misses targets that the proposed model recovers. The same images also show tighter concentration on target regions in both brighter and dimmer scenes, with less response spilling into nearby background.

Crowded cases make the difference clearer. Once targets lie close together, the baseline produces more misses, duplicate boxes, and stray positives, whereas Improved RF-DETR retains more true objects and suppresses redundant detections. The gain is not limited to stronger target emphasis; the examples also suggest cleaner separation between adjacent instances and surrounding clutter, which is consistent with the intended interaction among frequency reweighting, multi-scale refinement, and query recalibration.

Figure 8 presents the feature response heatmap comparison designed for the baseline RF-DETR, the three two-module variants, and Improved RF-DETR. In the updated layout, the first column lists the input/model settings from top to bottom, and the remaining columns provide representative heatmap examples. This visualization is used to examine whether the complete model produces more concentrated responses on debris regions while suppressing background activations more effectively than partial module combinations.

Figure 9 further compares YOLOv11n, RT-DETR, YOLO26n, YOLO26s, and Improved RF-DETR on representative samples. The updated figure arranges the original image and detector results from top to bottom in the order Original, YOLOv11n, RT-DETR, YOLO26n, YOLO26s, and Improved RF-DETR, rather than from

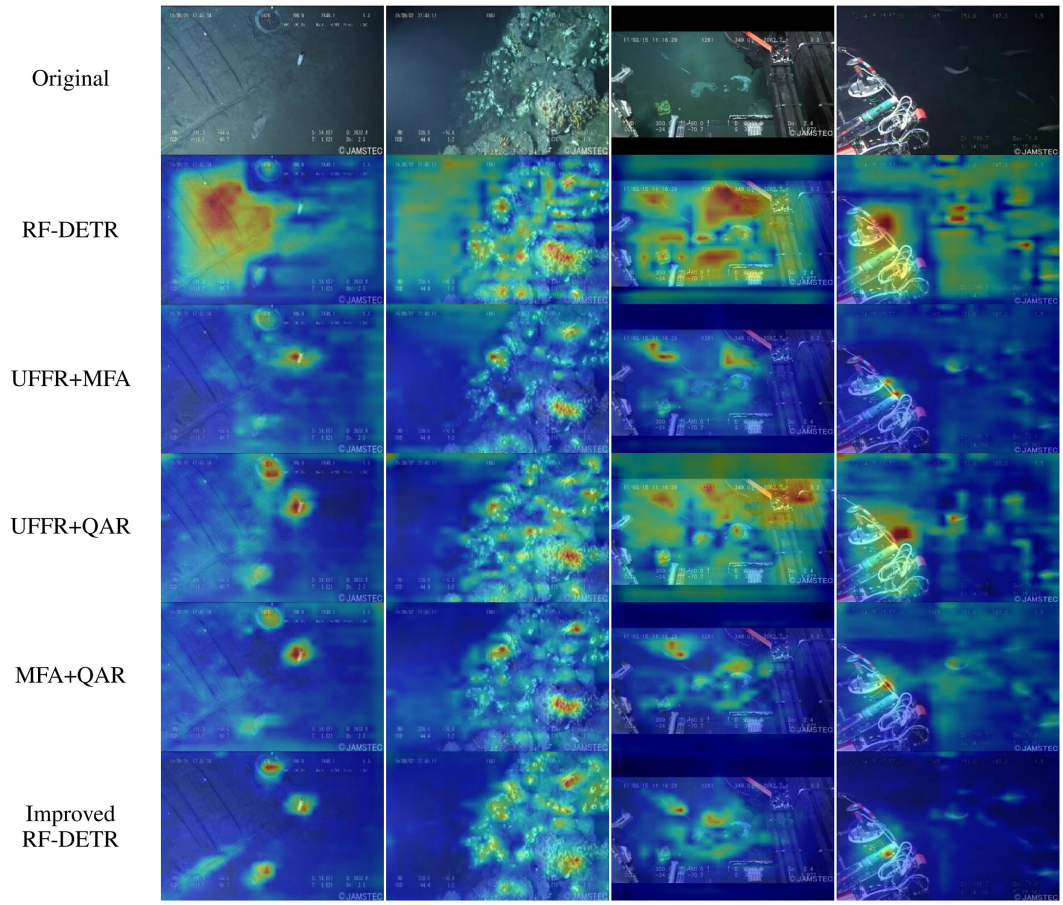


FIGURE 8
Feature response heatmap comparison among the baseline RF-DETR, two-module variants, and Improved RF-DETR. The first column lists the input/model settings from top to bottom, and the four remaining columns show representative heatmap examples from left to right. Warmer colors indicate stronger feature responses. Source: Based on data from Hong et al. (2020).

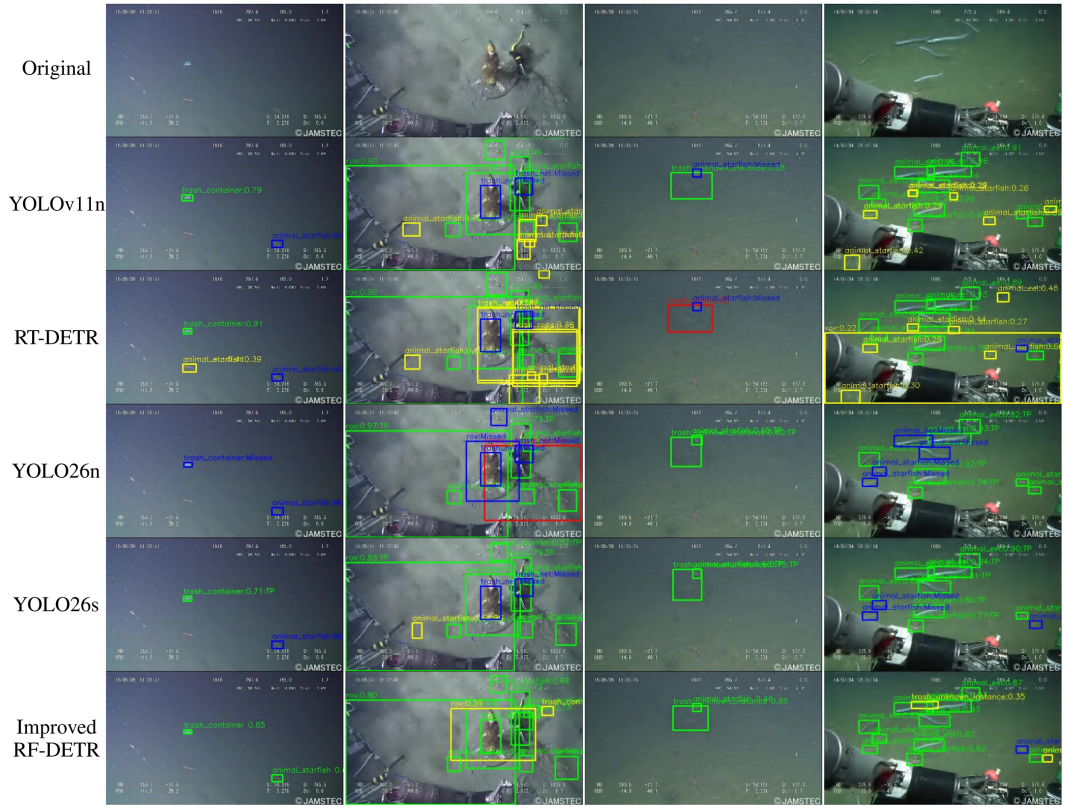


FIGURE 9
Comparison of representative detection results across five detectors on the TrashCan dataset. The first column lists the original image and detectors from top to bottom, and the four remaining columns show the corresponding representative examples from left to right. Source: Based on data from Hong et al. (2020).

left to right. YOLOv11n and YOLO26n more often miss adjacent or low-contrast targets, with YOLOv11n also showing stronger background confusion when objects are crowded or partly occluded. RT-DETR more often produces duplicate boxes or unstable boundaries, whereas YOLO26s is more balanced but still leaves some ambiguous small instances unresolved.

Improved RF-DETR keeps the most consistent compromise among coverage, localization, and duplicate suppression across these examples. In the first and third groups it better retains small neighboring targets, while in the second and fourth groups occasional duplicate detections are still visible. These remaining errors indicate that dense layouts, confused boundaries, and clutter continue to challenge all models, but the qualitative behavior of the proposed detector is still the most stable overall.

5.5 Category-wise performance analysis

Table 5 lists category-wise Precision, Recall, mAP@50, and mAP@50:95 for Improved RF-DETR on TrashCan, and Figure 10 extends the comparison to six baseline detectors: YOLO26n, RT-DETR, YOLOv11n, YOLOv8n, YOLO26s, and RF-DETR. Panels (A)–(D) correspond to Precision, mAP@50, mAP@50:95, and Recall, respectively.

Several categories show unusually strong values in Table 5. trash clothing, trash bottle, trash snack wrapper, and trash cup all reach 100% Precision and 100% Recall, and their mAP@50:95 values are 86.29%, 89.68%, 86.2%, and 75.17%, respectively. These results should be interpreted together with the category split statistics: trash clothing has 64 training, 11 validation, and 7 test instances (82 total); trash bottle has 79, 25, and 22 instances (126 total); trash snack wrapper has 56, 16, and 12 instances (84 total); and trash cup has 40, 15, and only 4 test instances (59 total). These four categories therefore have limited total sample sizes, and the final test subsets are even smaller. As a result, a small number of correctly localized test instances can produce exceptionally high Precision and Recall, so the apparently excellent values for these categories should be treated as sample-size-sensitive results rather than as evidence that these categories are always easy.

Lower scores cluster around categories that are thin, deformable, or visually entangled with the background. animal shells and animal etc fall to 69% and 66% Recall, trash tarp and trash net stay at 72% Recall, and mAP@50:95 drops to 43.86% for trash rope, 51.18% for animal starfish, and 53.27% for animal shells. These are precisely the classes for which uncertain boundaries and seafloor resemblance make both localization and classification harder.

The ordering in Figure 10 changes substantially with both metrics and categories, so the strongest detector for one class or threshold is not always strongest for another. Panels (A), (B), and (D) are

TABLE 5 Category-wise performance of Improved RF-DETR on the TrashCan dataset.

Class	Precision (%)	Recall (%)	mAP@50 (%)	mAP@50:95 (%)
rov	98.53	92	96.65	84.84
plant	93.18	91	89.42	56.82
animal fish	100	93	97.27	72.13
animal starfish	85.71	81	87.77	51.18
animal shells	85.37	69	86.84	53.27
animal crab	85.19	88	87.36	63.84
animal eel	89.47	83	88.77	61.99
animal etc	78.57	66	82.74	57.77
trash clothing	100	100	100	86.29
trash pipe	100	91	91.09	79.76
trash bottle	100	100	99.78	89.68
trash bag	95.59	89	92.24	73.45
trash snack wrapper	100	100	100	86.2
trash can	100	93	98.56	74.78
trash cup	100	100	100	75.17
trash container	100	96	99.46	81.42
trash unknown instance	93.61	88	89.89	63.98
trash branch	92.31	88	95.18	78.77
trash wreckage	85.71	84	90.88	68.22
trash tarp	100	72	83.47	71.37
trash rope	81.82	81	76.53	43.86
trash net	83.33	72	86.47	63.38
all	93.11	81	91.84	69.92

Source: Based on data from Hong et al. (2020).

compressed comparatively for easier categories, when panels (C) show a broad spread across models, which indicates that mAP@50:95 localization remains the main source of divergence. Panel (C) shows that under the stricter mAP@50:95 metric, the performance gaps among different models become more pronounced. The proposed method achieves the highest value in many categories, but the class-wise differences also show that category frequency, test-set size, object morphology, and background confusion all affect the final metric values. The color scheme in Figure 10 follows the same ordered logic: the model appearing most often at the bottom of the bar ranking is shown with the darkest tone, whereas the model appearing most often at the top is shown with the lightest tone. When several models share the same value, the corresponding color interval is divided evenly so that tied models remain visually separable.

6 Discussion

This study focuses on reliable underwater plastic debris detection in scenes where image degradation, weak texture, cluttered backgrounds, and irregular target shapes make local appearance cues unstable. The proposed Improved RF-DETR is designed around this requirement rather than around minimum model size alone. By combining underwater frequency-aware feature reweighting, multi-scale frequency-aware attention, and query adaptive reweighting, the model strengthens degraded visual cues, improves feature discrimination, and adjusts query responses for more stable classification and localization. The experimental results support the effectiveness of this combined design and indicate that the added modules contribute complementary improvements rather than simply increasing model capacity.

From an application perspective, the main value of the method is its balance between detection quality and computational cost. Underwater monitoring, deep-sea inspection, and robot-assisted surveys often rely on limited image acquisition opportunities, so a missed or unreliable single-frame prediction can be more costly than a moderate increase in computation. In this context, Improved RF-DETR is not intended to replace extremely lightweight detectors in scenarios where maximum frame rate or deployment on very small devices is the only priority. Instead, it provides a stronger option for settings where single-image reliability, reduced false alarms, and accurate localization are more important. The results also show that difficult cases remain: thin, deformable, partly occluded, or background-like targets are still harder to localize tightly, and some categories with very high scores have small test subsets, meaning that their apparent performance should be interpreted cautiously.

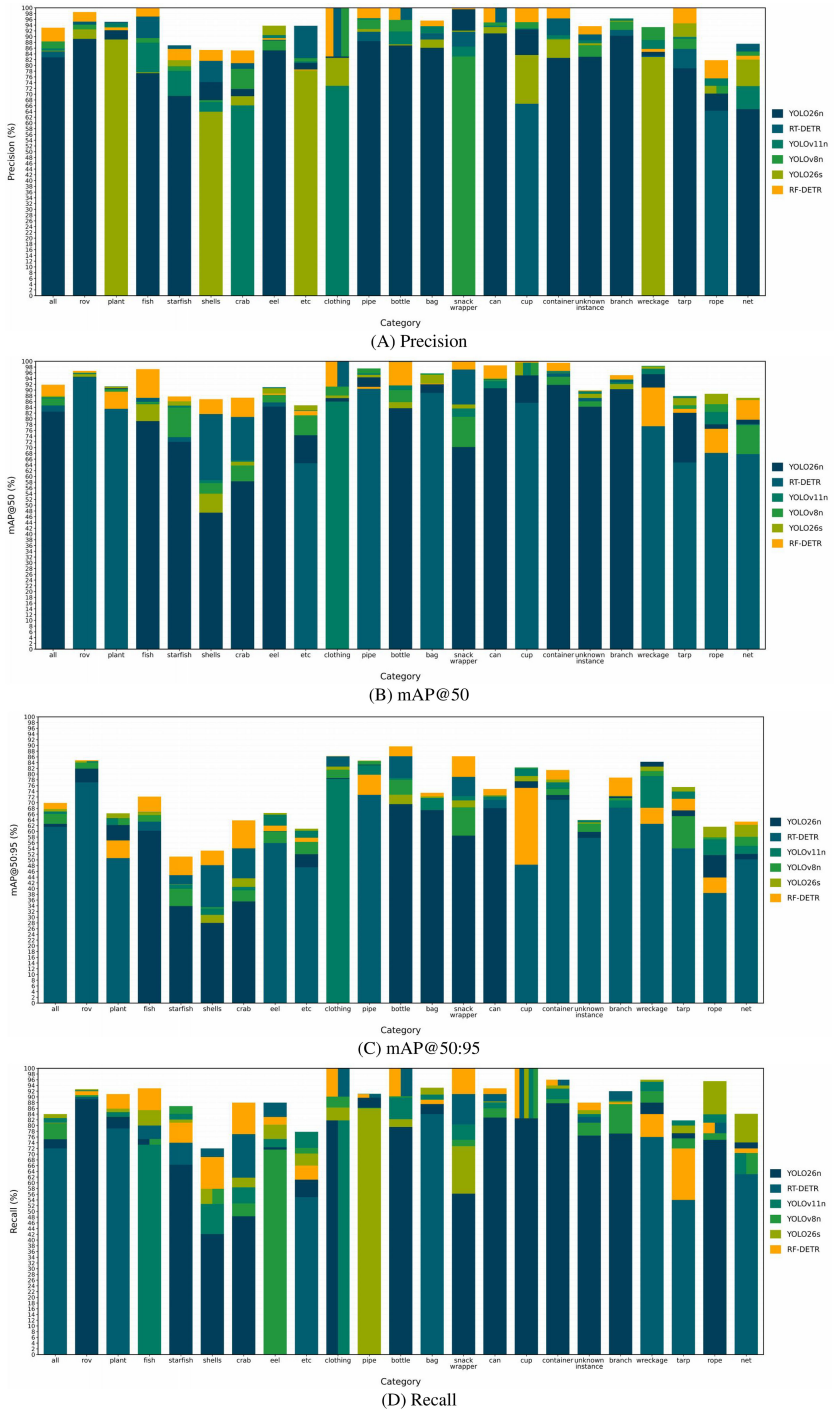


FIGURE 10
Category-wise comparison of six detectors on the TrashCan dataset. **(A–D)** compare Precision, mAP@50, mAP@50:95, and Recall, respectively, for YOLO26n, RT-DETR, YOLOv11n, YOLOv8n, YOLO26s, and RF-DETR. The colors follow an ordered dark-to-light scheme: the model appearing most often at the bottom of the bar ordering is shown with the darkest shade, whereas the model appearing most often at the top is shown with the lightest shade. If multiple models share the same value, the corresponding color interval is divided evenly so that tied cases remain distinguishable. **(C, D)** compare mAP@50:95 and Recall, respectively, using the same detector ordering and color scheme as **(A, B)**. Source: Based on data from Hong et al. (2020).

Several limitations should therefore be noted. First, the evaluation is still based on a limited number of underwater debris datasets, and the DeepTrash experiment involves transfer fine-tuning rather than zero-shot generalization. More diverse underwater sites, imaging devices, turbidity levels, lighting conditions, and debris distributions are needed before the robustness of the method can be fully established. Second, the present study mainly evaluates static images, while practical underwater systems often collect sequential observations. Temporal consistency, multi-frame fusion, and video-based uncertainty reduction may help reduce missed detections and duplicate predictions. Third, the comparison set contains many lightweight or small-capacity detectors and only

a limited number of medium- or high-capacity baselines, so the current results do not fully answer how the method compares with a broader range of stronger non-lightweight models.

Future research will address these limitations in several directions. More medium-sized and large nonlightweight models will be added to the benchmark, including larger YOLO variants, stronger DETR-family detectors, two-stage detectors, and other high-capacity architectures, so that the proposed method can be evaluated against models with comparable or greater representational capacity. In addition, future work will expand cross-domain testing to more underwater environments, investigate stronger adaptation strategies, incorporate temporal information from underwater video, and improve boundary-aware localization for thin or deformable debris. Larger and more balanced category-level test sets will also be important for determining whether high scores on small-sample categories remain stable under broader evaluation.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material. Further inquiries can be directed to the corresponding author.

Author contributions

YL: Validation, Visualization, Writing – original draft, Writing – review & editing. ML: Resources, Supervision, Writing – review & editing. HPY: Data curation, Investigation, Software, Writing – review & editing. HYY: Conceptualization, Formal analysis, Methodology, Writing – review & editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

Barnard, K. B., Daniels, J., Roberts, P. L. D., Orenstein, E. C., Masmitja, I., Takahashi, J., et al. (2024). DeepSTARia: enabling autonomous, targeted observations of ocean life in the deep sea. *Front. Mar. Sci.* 11, 1357879. doi: 10.3389/fmars.2024.1357879

Chen, G., Mao, Z., Wang, K., and Shen, J. (2023). HTDet: A hybrid transformer-based approach for underwater small object detection. *Remote Sens.* 15, 1076. doi: 10.3390/rs15041076

Corrigan, B. C., Tay, Z. Y., and Konovessis, D. (2023). Real-time instance segmentation for detection of underwater litter as a plastic source. *J. Mar. Sci. Eng.* 11, 1532. doi: 10.3390/jmse11081532

Cui, J., Zhou, S., Xu, G., Liu, X., and Gao, X. (2025). Marine debris detection in real time: A lightweight UTNet model. *J. Mar. Sci. Eng.* 13, 1560. doi: 10.3390/jmse13081560

Deluxni, N., and Sudhakaran, P. (2025). Underwater debris detection using YOLOv12 with enhanced feature extraction using R-ELAN and FlashAttention network. *Results Eng.* 28, 107282. doi: 10.1016/j.rineng.2025.107282

Er, M. J., Chen, J., Zhang, Y., and Gao, W. (2023). Research challenges, recent advances, and popular datasets in deep learning-based underwater marine object detection: A review. *Sensors* 23, 1990. doi: 10.3390/s23041990

Feng, J., and Jin, T. (2024). CEH-YOLO: A composite enhanced YOLO-based model for underwater object detection. *Ecol. Inf.* 82, 102758. doi: 10.1016/j.ecoinf.2024.102758

Gao, Z., Shi, Y., and Li, S. (2024). Self-attention and long-range relationship capture network for underwater object detection. *J. King Saud Univ. Comput. Inf. Sci.* 36, 101971. doi: 10.1016/j.jksuci.2024.101971

Guo, L., Liu, X., Ye, D., He, X., Xia, J., and Song, W. (2025). Underwater object detection algorithm integrating image enhancement and deformable convolution. *Ecol. Inf.* 89, 103185. doi: 10.1016/j.ecoinf.2025.103185

Hong, J., Fulton, M., and Sattar, J. (2020). TrashCan: A semantically-segmented dataset towards visual detection of marine debris. *arXiv*. doi: 10.48550/arXiv.2007.08097

Jian, M., Yang, N., Tao, C., Zhi, H., and Luo, H. (2024). Underwater object detection and datasets: A survey. *Intelligent Mar. Technol. Syst.* 2, 9. doi: 10.1007/s44295-024-00023-6

Lakshmanan, B., Selvakumar, B., Nivetika, R., and Karpagalakshmi, R. (2025). MarineDetNet: Enhancing underwater image quality for improved marine object detection. *Iranian J. Sci. Technology Trans. Electrical Eng.* 49, 1939–1957. doi: 10.1007/s40998-025-00834-1

Li, S., and Peng, X. (2025). DyAqua-YOLO: A high-precision real-time underwater object detection model based on dynamic adaptive architecture. *Front. Mar. Sci.* 12, 1678417. doi: 10.3389/fmars.2025.1678417

Li, T., Gang, Y., Li, S., and Shang, Y. (2025a). A small underwater object detection model with enhanced feature extraction and fusion. *Sci. Rep.* 15, 2396. doi: 10.1038/s41598-025-85961-9

Li, X., Wang, Y., Zhao, Y., and Chen, G. (2024). UW-DETR: Feature fusion enhanced RT-DETR for improving underwater object detection. *IEEE Access* 12, 191967–191979. doi: 10.1109/ACCESS.2024.3515960

Li, X., Zhao, Y., Su, H., Wang, Y., and Chen, G. (2025b). Efficient underwater object detection based on feature enhancement and attention detection head. *Sci. Rep.* 15, 5973. doi: 10.1038/s41598-025-89421-2

Li, Z., Kovachki, N. B., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., et al (2021). "Fourier neural operator for parametric partial differential equations," in *Proceedings of the Ninth International Conference on Learning Representations (ICLR 2021)* (Virtual: OpenReview.net).

Lin, C., Zhang, Y., and Chen, G. (2026). Lightweight underwater debris detection model based on improved RT-DETR. *Mar. pollut. Bull.* 222, 118537. doi: 10.1016/j.marpolbul.2025.118537

Liu, J., Chen, H., Li, Z., and Gu, H. (2024). Multi-scale frequency-spatial domain attention fusion network for building extraction in remote sensing images. *Electronics* 13, 4642. doi: 10.3390/electronics13234642

Liu, M., Wu, Y., Li, R., and Lin, C. (2025). LFN-YOLO: Precision underwater small object detection via a lightweight reparameterized approach. *Front. Mar. Sci.* 11, 1513740. doi: 10.3389/fmars.2024.1513740

Liu, S., Li, F., Zhang, H., Yang, X., Qi, X., Su, H., et al (2022). "DAB-DETR: Dynamic anchor boxes are better queries for DETR," in *Proceedings of the Tenth International Conference on Learning Representations (ICLR 2022)* (Virtual: OpenReview.net).

Liu, T., Wang, Q., Peng, Z., Jiang, G., and Chen, F. (2026). Enhancement of underwater image with multi-distortion for object detection. *Circuits Syst. Signal Process.* 45, 1018–1047. doi: 10.1007/s00034-025-03220-8

Madricardo, F., Ghezzo, M., Nesto, N., Mc Kiver, W. J., Faussone, G. C., Fiorin, R., et al. (2020). How to deal with seafloor marine litter: An overview of the state-of-the-art and future perspectives. *Front. Mar. Sci.* 7, 505134. doi: 10.3389/fmars.2020.505134

Maximenko, N., Corradi, P., Law, K. L., Van Sebille, E., Garaba, S. P., Lampitt, R. S., et al. (2019). Toward the integrated marine debris observing system. *Front. Mar. Sci.* 6, 447. doi: 10.3389/fmars.2019.00447

Meng, D., Chen, X., Fan, Z., Zeng, G., Li, H., Yuan, Y., et al (2021). "Conditional DETR for fast training convergence," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV 2021)* (Piscataway, NJ: IEEE), 3651–3660. doi: 10.1109/ICCV48922.2021.00363

Prakash, N., and Zielinski, O. (2025). AI-enhanced real-time monitoring of marine pollution: part 1-a state-of-the-art and scoping review. *Front. Mar. Sci.* 12, 1486615. doi: 10.3389/fmars.2025.1486615

Qin, Z., Zhang, P., Wu, F., and Li, X. (2021). "FcaNet: Frequency channel attention networks," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV 2021)* (Piscataway, NJ: IEEE), 783–792. doi: 10.1109/ICCV48922.2021.00082

Rao, Y., Zhao, W., Zhu, Z., Lu, J., and Zhou, J. (2021). "Global filter networks for image classification," in *Advances in Neural Information Processing Systems*, Vol. 34 (Red Hook, NY: Curran Associates, Inc.), 980–993.

Robinson, I., Robicheaux, P., Popov, M., Ramanan, D., and Peri, N. (2026). "RF-DETR: Neural architecture search for real-time detection transformers," in *Proceedings of the Fourteenth International Conference on Learning Representations (ICLR 2026)* (Rio de Janeiro: OpenReview.net).

Rohan, A., Tolie, H. F., Hasan, M. J., and Kannan, S. (2025). Full autonomy in underwater robotics systems: A realistic prospect? *Eng. Appl. Artif. Intell.* 162, 112638. doi: 10.1016/j.engappai.2025.112638

Tata, G., Royer, S.-J., Poirion, O., and Lowe, J. (2021). A robotic approach towards quantifying epipelagic bound plastic using deep visual models. *arXiv*. doi: 10.48550/arXiv.2105.01882

Wang, W., Yu, Z., and Huang, M. (2025). Refining features for underwater object detection at the frequency level. *Front. Mar. Sci.* 12, 1544839. doi: 10.3389/fmars.2025.1544839

Wang, Y., Zhang, X., Yang, T., and Sun, J. (2022). "Anchor DETR: Query design for transformer-based detector," in *Proceedings of the Thirty-Sixth AAAI Conference on Artificial Intelligence*, Vol. 36 (Palo Alto, CA: AAAI Press), 2567–2575. doi: 10.1609/aaai.v36i3.20158

Xie, H., Xin, G., Kang, S., and Xie, Q. (2025). "Efficient-Deformable-DETR: Enhancing underwater small target detection in complex environments," in *E3S Web of Conferences*, Vol. 603 (Les Ulis: EDP Sciences), 04014. doi: 10.1051/e3sconf/202560304014

Yan, S., Yang, B., Chen, A., Zhao, X., and Zhang, S. (2025). Multi-scale convolutional attention frequency-enhanced transformer network for medical image segmentation. *Inf. Fusion* 119, 103019. doi: 10.1016/j.inffus.2025.103019

Ying, S., Song, X., and Wang, H. (2025). High-frequency-based multi-spectral attention for domain generalization. *Artif. Intell. Rev.* 58, 245. doi: 10.1007/s10462-025-11217-7

Yuanwei, J., Yijiang, D., Xuemei, W., and Khairuddin, A. S. M. (2026). An improved YOLOv11 network for marine debris detection in underwater environment. *Sci. Rep.* 16, 7074. doi: 10.1038/s41598-026-38305-0

Zhang, D., Zhang, J., Wei, T., He, Z., and Lin, Z. (2025). Underwater small object detection via automatic encoding pyramid with neighborhood information. *J. Ocean Eng. Mar. Energy* 11, 841–854. doi: 10.1007/s40722-025-00405-w